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Inventor(s)

ONISHI; Hideaki

INKJET PRINTING APPARATUS

Abstract

The input unit receives a first command for switching a first ink discharge surface from a capped state to a decapped state and outputs a first input signal, and receives a second command for switching a second ink discharge surface from a capped state to a decapped state and outputs a second input signal. The control unit can perform step a) of switching the first ink discharge surface from a capped state to a decapped state when the first input signal is received and it is detected that a first head unit has moved from a first driving position to a first maintenance position, and step b) of switching the second ink discharge surface from a capped state to a decapped state when the second input signal is received and it is detected that a second head unit has moved from a second driving position to a second maintenance position.

Inventors: ONISHI; Hideaki (Kyoto-shi, JP)

Applicant: SCREEN Holdings Co., Ltd. (Kyoto, JP)

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims the benefit of Japanese application No. 2024-037588, filed on 11 Mar. 2024, the disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an inkjet printing apparatus that performs printing on a printing medium such as paper by an inkjet method.

Description of the Background Art

[0003] Conventionally, in an inkjet printing apparatus, various kinds of maintenance are performed in order to prevent clogging of a nozzle that discharges ink. For example, in order to prevent a nozzle from being dried during stop of printing, the nozzle is covered with a cap, whereby an ink discharge surface including the nozzle is isolated and enclosed from an outside atmosphere. Such a method of performing maintenance of a nozzle is described in, for example, Japanese Patent Application Laid-Open No. 2014-136396.

[0004] Japanese Patent Application Laid-Open No. 2014-136396 discloses a maintenance unit (9) for performing maintenance of an image forming apparatus. The maintenance unit (9) includes a cap (91) for capping a nozzle surface (124), a wiper member (94) for wiping the nozzle surface (124), a suction pump (97) connected to the cap (91), and a waste tank (98) connected to the suction pump (97) (paragraphs and [0039]). Meanwhile, the maintenance unit (9) is provided so as to be movable between a maintenance position where the nozzle surface (124) is capped with the cap (91) and a receding position where the maintenance unit (9) recedes from the maintenance position (paragraph [0043], FIG. 5(a), and FIG. 5(b)).

[0005] In recent years, an apparatus that performs an operation (capping) of covering an ink discharge surface including a nozzle with a cap and isolating and enclosing the ink discharge surface from an outside atmosphere, and an operation (decapping) of separating the cap from the ink discharge surface and exposing the ink discharge surface to the outside atmosphere, in response to an input operation performed on an input screen by an operator or the like, is used in many cases. In the case of such an apparatus, the operator designates a head to be an object on the input screen, and inputs an instruction for “capping” or “decapping”, to thereby automatically perform capping of the designated head.

[0006] In this regard, in order to prevent the nozzle from being dried during stop of printing, it is preferred to enclose the ink discharge surface including the nozzle with the cap as long as possible while the nozzle is not subjected to any process such as wiping. That is, it is preferred to keep the ink discharge surface in a “capped state” as long as possible while the nozzle is not subjected to any process such as wiping. To this end, it is preferred that the operator designate a head including the nozzle and input an instruction for “decapping” on the input screen immediately before wiping or the like of the nozzle. That is, it is preferred that the operator designate a head and input an instruction for “decapping” on the input screen. However, this necessitates a time and effort of the operator appearing in person in front of the input screen and performing an input operation for each change of a head to be subjected to maintenance, increasing a burden on the operator.

SUMMARY OF THE INVENTION

[0007] It is an object of the present invention to provide a technology that can reduce a burden on an operator while keeping an ink discharge surface including a nozzle in a capped state as long as possible in sequentially performing maintenance of a plurality of heads including a plurality of nozzles.

[0008] To solve the above-described problem, the first invention of the present application is

directed to an inkjet printing apparatus that performs printing on a printing medium by an inkjet method and includes an apparatus casing, a first head unit, a second head unit, a first slider, a second slider, a first detection unit, a second detection unit, an input unit, and a control unit. The first head unit includes a first discharge head having a first ink discharge surface in which a first discharge port facing down, through which first ink is discharged, is formed, a first cap configured to cap the first ink discharge surface, and a first movement mechanism configured to switch the first ink discharge surface between a capped state in which the first ink discharge surface is isolated from an outside atmosphere with the first cap and a decapped state in which the capped state is canceled. The second head unit includes a second discharge head having a second discharge surface in which a second discharge port facing down, through which second ink is discharged, is formed, a second cap configured to cap the second ink discharge surface, and a second movement mechanism configured to switch the second ink discharge surface between a capped state in which the second ink discharge surface is isolated from the outside atmosphere with the second cap and a decapped state in which the capped state is canceled. The first slider supports the first head unit such that the first head unit is slidable between a first driving position in an inside space of the apparatus casing and a first maintenance position in an outside space of the apparatus casing. The second slider is provided separately from the first slider and supports the second head unit such that the second head unit is slidable between a second driving position in the inside space of the apparatus casing and a second maintenance position in the outside space of the apparatus casing. The first detection unit is configured to detect movement of the first head unit between the first driving position and the first maintenance position and output a first detection signal. The second detection unit is configured to detect movement of the second head unit between the second driving position and the second maintenance position and output a second detection signal. The input unit is configured to receive a first command including a command for switching the first ink discharge surface from the capped state to the decapped state and output a first input signal upon receipt of the first command, the input unit being configured to receive a second command including a command for switching the second ink discharge surface from the capped state to the decapped state and output a second input signal upon receipt of the second command. The control unit is electrically connected to the first discharge head, the first movement mechanism, the first detection unit, the second discharge head, the second movement mechanism, the second detection unit, and the input unit. The control unit is capable of individually performing a step a) of controlling the first movement mechanism so as to switch the first ink discharge surface from the capped state to the decapped state after elapse of a first predetermined time period when it is detected on the basis of the first detection signal received from the first detection unit, that the first head unit has moved from the first driving position to the first maintenance position while the first input signal has been received from the input unit, and a step b) of controlling the second movement mechanism so as to switch the second ink discharge surface from the capped state to the decapped state after elapse of a second predetermined time period when it is detected on the basis of the second detection signal received from the second detection unit, that the second head unit has moved from the second driving position to the second maintenance position while the second input signal has been received from the input unit.

[0009] The second invention of the present application is directed to the inkjet printing apparatus according to the first invention, wherein after the step a), the control unit controls the first movement mechanism so as to switch the first ink discharge surface from the decapped state to the capped state when it is detected on the basis of the first detection signal further received from the first detection unit, that the first head unit has moved from the first maintenance position to the first driving position. Meanwhile, after the step b), the control unit controls the second movement mechanism so as to switch the second ink discharge surface from the decapped state to the capped state when it is detected on the basis of the second detection signal further received from the second detection unit, that the second head unit has moved from the second maintenance position

to the second driving position.

[0010] The third invention of the present application is directed to the inkjet printing apparatus according to the first invention or the second invention, the control unit recognizes presence of an error and waits while keeping the first ink discharge surface in the capped state when it is detected on the basis of the first detection signal received from the first detection unit, that the first head unit has moved from the first driving position to the first maintenance position while the first input signal is not received from the input unit. Meanwhile, the control unit recognizes presence of an error and waits while keeping the second ink discharge surface in the capped state when it is detected on the basis of the second detection signal received from the second detection unit, that the second head unit has moved from the second driving position to the second maintenance position while the second input signal is not received from the input unit.

[0011] The fourth invention of the present application is directed to the inkjet printing apparatus according to any of the first invention to the third invention, further including a first circulation path, a first circulation pump, a second circulation path, and a second circulation pump. The first circulation path includes a first supply tank in which the first ink supplied to the first discharge head is stored, a first collecting tank in which the first ink collected from the first discharge head is stored, and a first feedback pipe connecting the first collecting tank and the first supply tank. The first circulation pump is interposed in the first feedback pipe and is configured to deliver the first ink from the first collecting tank to the first supply tank via the first feedback pipe. The second circulation path includes a second supply tank in which the second ink supplied to the second discharge head is stored, a second collecting tank in which the second ink collected from the second discharge head is stored, and a second feedback pipe connecting the second collecting tank and the second supply tank. The second circulation pump is interposed in the second feedback pipe and is configured to deliver the second ink from the second collecting tank to the second supply tank via the second feedback pipe. The control unit is further electrically connected to the first circulation pump and the second circulation pump. Further, the control unit stops driving the first circulation pump when it is detected that the first head unit has moved from the first driving position to the first maintenance position. Meanwhile, the control unit stops driving the second circulation pump when it is detected that the second head unit has moved from the second driving position to the second maintenance position.

[0012] According to the first invention to the fourth invention of the present application, the ink discharge surface of the discharge head of the head unit to be subjected to maintenance is switched to a “decapped state” immediately before the maintenance of each head unit, and thus a time period for which the ink discharge surface of the head unit to be subjected to the maintenance is exposed to the outside atmosphere can be shortened. Consequently, each discharge port can be prevented from being dried. Further, commands for switching a plurality of head units to a “decapped state” can be input at once, which reduces a burden on the operator.

[0013] Especially, according to the second invention of the present application, immediately after maintenance is finished and the head unit is returned back to the inside space of the apparatus casing, the ink discharge surface of the returned head unit can be switched again to a “capped state”. Thus, a time period for which each ink discharge surface is exposed to the outside atmosphere can be further shortened. Therefore, each ink discharge surface can be further prevented from being dried.

[0014] Especially, according to the third invention of the present application, in a case in which a head unit not scheduled to be pulled out to the outside space is pulled out, the presence of an error can be recognized, and then the rest of work can be discontinued.

[0015] Especially, according to the fourth invention of the present application, circulation of ink is stopped, and thus ink leakage from the discharge port of the pulled-out head unit, intake of air through the discharge port, and the like can be prevented even though the head unit has been pulled out to the outside space. This suppresses a change in a pressure applied to the discharge head, to

thereby suppress occurrence of damage or the like. Further, circulation of ink is stopped immediately before maintenance of each head unit, and thus the discharge port can be further prevented from being dried.

[0016] These and other objects, features, aspects, and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a view conceptually showing a configuration of an inkjet printing apparatus;
[0018] FIG. 2 is a perspective view diagrammatically showing a configuration of a head unit;
[0019] FIG. 3 is a view conceptually showing configurations of a discharge head and an ink supply unit of the head unit;
[0020] FIG. 4 is a bottom view of the discharge head and a head holding plate of the head unit as seen from below;
[0021] FIG. 5 is a view conceptually showing a positional relationship between the head unit and an apparatus casing of the inkjet printing apparatus;
[0022] FIG. 6 is a side view diagrammatically showing configurations of the head unit and a slider;
[0023] FIG. 7 is a side view diagrammatically showing configurations of the head unit and the slider;
[0024] FIG. 8 is a block diagram showing connection between a control unit and each component of the inkjet printing apparatus; and
[0025] FIG. 9 is a flowchart showing a procedure for performing maintenance of the head unit.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0026] Hereinafter, a preferred embodiment of the present invention will be described with reference to the drawings. Note that components described in this preferred embodiment are mere examples and are not intended to limit the scope of the present invention to those only. In the drawings, for the purpose of easier understanding, the dimensions or the number of respective components are overstated or understated in some portions of illustration, as necessary.

1. Configuration of Inkjet Printing Apparatus

[0027] FIG. 1 is a view conceptually showing a configuration of an inkjet printing apparatus 1 according to one preferred embodiment of the present invention. The inkjet printing apparatus 1 is an inkjet printing machine that discharges droplets of water-based ink toward continuous paper 10 in a shape of a long strip from a plurality of head units 35 while conveying the continuous paper 10, to record characters or images on a surface of the continuous paper 10. Meanwhile, the continuous paper 10 in a shape of a long strip is just one example of a printing medium. The printing medium may be a cut sheet, a plastic film, a cardboard, metal foil, a material made of glass, or the like. In other words, the inkjet printing apparatus 1 may be any apparatus that can perform printing on a printing medium by an inkjet method.

[0028] The inkjet printing apparatus 1 includes a conveyor unit 2, a printing unit 3, an apparatus casing 4, a control unit 9, and an input unit 11. Further, as later described in detail, the printing unit 3 includes a plurality of head units 35. In the present preferred embodiment, the printing unit 3 includes four head units 35. The four head units 35 have substantially the same configuration with each other. Hence, in the following description and the drawings, when it is necessary to distinguish the four head units 35, suffix characters “a” to “d” are added to the reference numeral “35”, and the four head units 35 are referred to as a “first head unit 35a”, a “second head unit 35b”, a “third head unit 35c”, and a “fourth head unit 35d”. Meanwhile, when it is unnecessary to distinguish the four head units 35, the suffix characters are omitted.

[0029] Further, in the inkjet printing apparatus **1**, one ink supply unit **36**, one slider **41** (refer to FIG. 5), and one detection unit (refer to FIG. 5) that correspond to each of the four head units **35** are provided. The ink supply units **36** corresponding to the four head units **35**, respectively, have substantially the same configuration with each other. Hence, in the following description and the drawings, when it is necessary to distinguish the four ink supply units **36**, suffix characters “a” to “d” are added to the reference numeral “**36**”, and the four ink supply units **36** are referred to as a “first ink supply unit **36a**”, a “second ink supply unit **36b**”, a “third ink supply unit **36c**”, and a “fourth ink supply unit **36d**”. Meanwhile, when it is unnecessary to distinguish the four ink supply units **36**, the suffix characters are omitted. Moreover, the sliders **41** corresponding to the four head units **35**, respectively, have substantially the same configuration with each other. Hence, in the following description and the drawings, when it is necessary to distinguish the four sliders **41**, suffix characters “a” to “d” are added to the reference numeral “**41**”, and the four sliders **41** are referred to as a “first slider **41a**”, a “second slider **41b**”, a “third slider **41c**”, and a “fourth slider **41d**”. Meanwhile, when it is unnecessary to distinguish the four sliders **41**, the suffix characters are omitted. Furthermore, the detection units **45** corresponding to the four head units **35**, respectively, have substantially the same configuration with each other. Hence, in the following description and the drawings, when it is necessary to distinguish the four detection units **45**, suffix characters “a” to “d” are added to the reference numeral “**45**”, and the four detection units **45** are referred to as a “first detection unit **45a**”, a “second detection unit **45b**”, a “third detection unit **45c**”, and a “fourth detection unit **45d**”. Meanwhile, when it is unnecessary to distinguish the four detection units **45**, the suffix characters are omitted.

[0030] The conveyor unit **2** is a mechanism configured to convey the continuous paper **10** along a predetermined conveying path in a conveying direction extending along a length direction of the continuous paper **10**. The continuous paper **10** is stretched over a plurality of conveyor rollers **12**. The continuous paper **10** is conveyed along the conveying path formed by the plurality of conveyor rollers **12**. Each of the conveyor rollers **12** rotates about an axis extending in a direction (width direction) perpendicular to the conveying direction, to thereby guide the continuous paper **10** to a downstream side in the conveying path. Further, the continuous paper **10** is under tension in the conveying direction. This reduces slack or wrinkles in the continuous paper **10** during conveying.

[0031] The printing unit **3** includes the plurality of head units **35** and the plurality of ink supply units **36**. In the present preferred embodiment, the printing unit **3** includes four head units **35** and four ink supply units **36**. As described above, the four head units **35** have substantially the same configuration with each other. Further, as described above, the four ink supply units **36** have substantially the same configuration with each other.

[0032] The four head units **35** are arranged while being spaced from each other along the conveying direction. Each of the four head units **35** discharges ink droplets toward a surface (upper surface) of the continuous paper **10** being conveyed in the conveying direction below the four head units **35**, from nozzles **103** (refer to FIG. 3 described later). In the present preferred embodiment, the four head units **35** discharge ink of different colors, respectively, to thereby each record a monochromatic image on the surface (upper surface) of the continuous paper **10**. For example, the four head units **35** discharge cyan ink, magenta ink, yellow ink, and black ink, respectively. Then, the four monochromatic images are superimposed, so that a multicolor image is formed on the upper surface of the continuous paper **10**.

[0033] Meanwhile, in the following description and the drawings, when it is necessary to distinguish ink of different colors discharged from the four head units **35**, cyan ink is referred to as a “first ink”, and the head unit **35** that discharges the first ink is referred to as the “first head unit **35a**”. Magenta ink is referred to as a “second ink”, and the head unit that discharges the second ink is referred to as the “second head unit **35b**”. Yellow ink is referred to as a “third ink”, and the head unit **35** that discharges the third ink is referred to as the “third head unit **35c**”. Black ink is referred to as the “fourth ink”, and the head unit **35** that discharges the fourth ink is referred to as the

“fourth head unit **35d**”. However, the colors of ink discharged from the respective head units **35a** to **35d** are not limited to those. The printing unit **3** may include a discharge head that discharges ink of a spot color such as orange, blue, or violet, in addition to cyan, magenta, yellow, and black. [0034] Further, the ink supply unit **36** connected to the “first head unit **35a**” is referred to as the “first ink supply unit **36a**”. The ink supply unit **36** connected to the “second head unit **35b**” is referred to as the “second ink supply unit **36b**”. The ink supply unit **36** connected to the “third head unit **35c**” is referred to as the “third ink supply unit **36c**”. The ink supply unit **36** connected to the “fourth head unit **35d**” is referred to as the “fourth ink supply unit **36d**”. The configurations thereof will be later described in detail.

[0035] FIG. **2** is a perspective view diagrammatically showing a configuration of one head unit **35**. Note that, in FIG. **2**, a guided portion **176** mounted onto a support stand **175** described later is omitted. As shown in FIG. **2**, each of the head units **35** includes a plurality of discharge heads **811**, a cap **812**, a main movement mechanism **813**, a head holding plate **814**, and a subsidiary movement mechanism **815**. In the present preferred embodiment, each of the head units **35** includes five discharge heads **811**.

[0036] First, a configuration of the discharge head **811** is described. FIG. **3** is a view conceptually showing configurations of the plurality of discharge heads **811** and the ink supply unit **36** included in one head unit **35**. Specifically, FIG. **3** is a view conceptually showing the configurations of the five discharge heads **811** and the ink supply unit **36** included in one head unit **35**. FIG. **4** is a bottom view of the five discharge heads **811** and the head holding plate **814** as seen from below. Meanwhile, the five discharge heads **811** have substantially the same configuration with each other. Hence, in FIG. **3**, only one of the five discharge heads **811** is shown in detail, and the other four discharge heads **811** are shown in a simplified manner.

[0037] As shown in FIG. **3**, each of the five discharge heads **811** includes a casing **101**, an internal tank **102**, and a plurality of nozzles **103**. The casing **101** forms an outer frame of the discharge head **811**. The internal tank **102** is provided in the casing **101**, and its corresponding ink can be temporarily stored therein. The plurality of nozzles **103** are arranged at regular intervals along the conveying direction and the width direction of the continuous paper on a lower side in the casing **101**. Each of the plurality of nozzles **103** faces the upper surface of the continuous paper **10** being conveyed therebelow. Further, each of the plurality of nozzles **103** communicates with the internal tank **102**. Moreover, each of the plurality of nozzles **103** includes a plurality of piezoelectric elements **104** serving as pressure generation elements, an ink chamber **105**, and a discharge port **151**. The ink chamber **105** communicates with the internal tank **102**.

[0038] During discharge of ink, ink flows down from the internal tank **102** to the ink chamber **105**. Then, under the control of the piezoelectric elements **104**, ink in the ink chamber **105** is pressurized, and thus is discharged in the form of liquid droplets through the discharge port **151**. That is, the discharge head **811** has an ink discharge surface **161** (lower surface) in which the discharge port **151** facing down, through which ink is discharged, is formed. Alternatively, the nozzle **103** may be a so-called thermal nozzle in which ink in the ink chamber **105** is heated to generate bubbles and thus is pressurized.

[0039] As shown in FIG. **2** and FIG. **4**, the five discharge heads **811** are arranged in a zigzag (staggered) pattern in an upper surface of the head holding plate **814**. The head holding plate **814** is a plate-shaped member extending in a rectangular shape and substantially horizontally. Hereinafter, a length direction of the head holding plate **814**, a width direction of the head holding plate **814**, and a direction perpendicular to the head holding plate **814** are referred to as an “X direction”, a “Y direction”, and a “Z direction”, respectively.

[0040] The head holding plate **814** includes five through holes **120**. Each of the five through holes **120** passes through the head holding plate **814** along the Z direction. When the head holding plate **814** in which each discharge head **811** is placed on the upper surface is seen from below along the Z direction, the through holes **120** surround the plurality of nozzles **103** included in one discharge

head **811** and are each slightly smaller than the ink discharge surface **161** of one discharge head **811**. Thus, the plurality of discharge ports **151** of each of the discharge heads **811** are exposed on a lower-surface side of the head holding plate **814** via the through hole **120**. As a result, ink can be discharged from each of the discharge heads **811** to the continuous paper **10** being conveyed therebelow via the through holes **120** of the head holding plate **814**.

[0041] The cap **812** is a member having an area larger than that of the five discharge heads **811** or the head holding plate **814** as seen along the Z direction. The cap **812** is used for, for example, covering the nozzles **103** from below and isolating and enclosing the ink discharge surfaces **161** from an outside atmosphere, to prevent drying during stop of printing. That is, the cap **812** can cap the ink discharge surfaces **161** including the plurality of discharge ports **151**. Note that a state in which the nozzles **103** of the five discharge heads **811** are isolated and enclosed from the outside atmosphere with the cap **812** as described above is referred to as a “capped state”. Meanwhile, as shown in FIG. 2, the cap **812** has a shape of substantially rectangular parallelepiped box and has a recess shape recessed downward. As a result of this, during purge, the cap **812** vertically faces the five discharge heads **811** and also can receive ink discharged (falling) from the five discharge heads **811**. That is, the cap **812** can receive ink falling from the discharge port **151** when positioned at a facing position P1 (refer to FIG. 2) where the cap **812** faces the ink discharge surfaces **161** of the discharge heads **811** from below.

[0042] The main movement mechanism **813** includes a pair of linear guides **171** and **172**, a ball screw **173**, and a cap motor **174**. The pair of linear guides **171** and **172**, the ball screw **173**, and the cap motor **174** are placed on a substantially horizontal upper surface of the support stand **175** having a substantially rectangular parallelepiped shape. Further, as shown in FIG. 2, the pair of linear guides **171** and **172** are each placed along the X direction.

[0043] The ball screw **173** extends along the X direction and is placed between the pair of linear guides **171** and **172**. Further, the ball screw **173** is connected to the cap motor **174** and rotates when driven by the cap motor **174**. The cap **812** is connected to the ball screw **173** via a connecting member not shown. The cap **812** moves as led by the pair of linear guides **171** and **172** along the X direction by rotation of the ball screw **173**. Thus, the main movement mechanism **813** can move the cap **812** along the X direction between the facing position P1 where the cap **812** faces the ink discharge surfaces **161** of the five discharge heads **811** from below and a receding position P2 away from the facing position P1. As a result of this, the main movement mechanism **813** can move the cap **812**, to cause the cap **812** to face the ink discharge surfaces **161** of the five discharge heads **811** from below, or to inhibit the cap **812** from facing the ink discharge surfaces **161** from below.

[0044] The subsidiary movement mechanism **815** is a mechanism that moves up and down (along the Z direction) the head holding plate **814** on which the discharge heads **811** are placed, by approximately a few centimeters to tens of centimeters. When the subsidiary movement mechanism **815** moves down the head holding plate **814** with the cap **812** positioned at the facing position P1, the head holding plate **814** comes into contact with the cap **812**. This places the ink discharge surfaces **161** of the discharge heads **811** into an enclosed state (capped state). Further, when the subsidiary movement mechanism **815** moves up the head holding plate **814**, the ink discharge surfaces **161** of the discharge heads **811** are released from a state in which the ink discharge surfaces **161** are enclosed with the cap **812**.

[0045] Note that a state in which a “capped state” in which the nozzles **103** of the five discharge heads **811** are isolated and enclosed from the outside atmosphere with the cap **812** as described above is cancelled is referred to as a “decapped state”. That is, the subsidiary movement mechanism **815** can switch the ink discharge surfaces **161** between a “capped state” in which the ink discharge surfaces **161** are isolated from the outside atmosphere with the cap **812** and a “decapped state” in which the “capped state” is canceled. Below, a specific configuration of the subsidiary movement mechanism **815** is described. As shown in FIG. 2, the subsidiary movement mechanism **815** includes a support plate **851**, a head motor **852**, a ball screw **853**, a pair of linear

guides (not shown), and a screwing member **854**.

[0046] The support plate **851** has a substantially rectangular parallelepiped shape. The support plate **851** is fixed to an upper surface of the support stand **175** and extends in a plate shape along a substantially vertical direction. The head motor **852** is fixed to the support plate **851**. Further, the ball screw **853** is fixed to a rotation shaft of the head motor **852**. The ball screw **853** extends along the Z direction. The ball screw **853** is placed between the pair of linear guides (not shown) extending along the Z direction. The screwing member **854** is fixed to the head holding plate **814** in which the five discharge heads **811** are placed. Further, the screwing member **854** includes a through hole. Around the through hole of the screwing member **854**, an internal thread is formed. The ball screw **853** is inserted into the through hole of the screwing member **854**. The ball screw **853** is screwed into the internal thread of the screwing member **854**.

[0047] When the ball screw **853** is driven to rotate by the head motor **852**, the screwing member **854** screwed into the ball screw **853** and the head holding plate **814** to which the screwing member **854** is fixed move as led by the pair of linear guides along the Z direction. Thus, the subsidiary movement mechanism **815** can simultaneously move up and down the five discharge heads **811** placed on the head holding plate **814**. Consequently, the subsidiary movement mechanism **815** can bring the respective ink discharge surfaces **161** of the five discharge heads **811** close to the cap **812** to place them into a “capped state”, or can separate the ink discharge surfaces **161** from the cap **812** to place them into a “decapped state”.

[0048] Next, configurations of the first slider **41a** to the fourth slider **41d** are described. FIG. 5 is a view conceptually showing a positional relationship between the first head unit **35a** to the fourth head unit **35d** and the apparatus casing **4** in the inkjet printing apparatus **1**. As shown in FIG. 5, in driving the inkjet printing apparatus **1**, the first head unit **35a** to the fourth head unit **35d** are arranged side by side in an inside space of the apparatus casing **4**. Further, the control unit **9** and the input unit **11** are provided near a wall portion of the apparatus casing **4** perpendicular to a row of a first maintenance position Poa to a fourth maintenance position Pod described later. The first head unit **35a** to the fourth head unit **35d** are each movable between the inside space and an outside space of the apparatus casing **4**.

[0049] Specifically, as shown in FIG. 5, the first slider **41a** that supports the first head unit **35a** such that the first head unit **35a** can slide between a first driving position Pia in the inside space of the apparatus casing **4** and the first maintenance position Poa in the outside space of the apparatus casing **4** is provided. Aside from the first slider **41a**, the second slider **41b** that supports the second head unit **35b** such that the second head unit **35b** can slide between a second driving position Pib in the inside space of the apparatus casing **4** and the second maintenance position Pob in the outside space of the apparatus casing **4** is provided. Aside from the first slider **41a** and the second slider **41b**, the third slider **41c** that supports the third head unit **35c** such that the third head unit **35c** can slide between a third driving position Pic in the inside space of the apparatus casing **4** and the third maintenance position Poc in the outside space of the apparatus casing **4** is provided. Aside from the first slider **41a** to the third slider **41c**, the fourth slider **41d** that supports the fourth head unit **35d** such that the fourth head unit **35d** can slide between a fourth driving position Pid in the inside space of the apparatus casing **4** and the fourth maintenance position Pod in the outside space of the apparatus casing **4** is provided.

[0050] The first slider **41a**, the second slider **41b**, the third slider **41c**, and the fourth slider **41d** have substantially the same configuration with each other. Hence, the configurations of the first slider **41a** and the second slider **41b** are described below as an example. FIG. 6 is a side view diagrammatically showing the configurations of the first head unit **35a** and the first slider **41a**. FIG. 7 is a side view diagrammatically showing the configurations of the second head unit **35b** and the second slider **41b**.

[0051] As shown in FIG. 6, the first slider **41a** is a guide rail extending from the first driving position Pia in the inside space of the apparatus casing **4** toward the first maintenance position Poa

in the outside space of the apparatus casing **4**. The first slider **41a** includes a pair of first rail members **40a** each extending along the X direction. When a first support stand **175a** corresponding to the above-described support stand **175** is placed on the first slider **41a**, a pair of first guided portions **176a** provided in a lower surface of the first support stand **175a** are engaged with the first rail members **40a** so as to be slidable along the X direction. Thus, the first support stand **175a** and the first head unit **35a** held by the first support stand **175a** are guided as led by the first slider **41a** along the X direction. This allows the operator to pull out the first support stand **175a** and the first head unit **35a** held by the first support stand **175a** from the first driving position Pia in the apparatus casing **4** to the first maintenance position Poa outside the apparatus casing **4**, and push them back again to the first driving position Pia.

[0052] Further, in the first slider **41a**, the first detection unit **45a** including, for example, a photo-microsensor is fixed. Moreover, in the first support stand **175a**, a sensor dog (not shown) for position detection is fixed. The first detection unit **45a** outputs a detection signal each time the first detection unit **45a** detects the sensor dog that passes in the vicinity as the first support stand **175a** moves along the X direction. Hereinafter, the detection signal output from the first detection unit **45a** is referred to as a “first detection signal”. That is, the first detection unit **45a** detects movement of the first head unit **35a** being supported by the first support stand **175a** between the first driving position Pia and the first maintenance position Poa, and outputs the first detection signal. Thus, the control unit **9** described later can grasp a position of the first head unit **35a** by referring to the first detection signal output from the first detection unit **45a**.

[0053] As shown in FIG. 7, the second slider **41b** is a guide rail extending from the second driving position Pib in the inside space of the apparatus casing **4** toward the second maintenance position Pob in the outside space of the apparatus casing **4**. The second slider **41b** includes a pair of second rail members **40b** each extending along the X direction. When a second support stand **175b** corresponding to the above-described support stand **175** is placed on the second slider **41b**, a pair of second guided portions **176b** provided in a lower surface of the second support stand **175b** are engaged with the second rail members **40b** so as to be slidable along the X direction. Thus, the second support stand **175b** and the second head unit **35b** held by the second support stand **175b** are guided as led by the second slider **41b** along the X direction. This allows the operator to pull out the second support stand **175b** and the second head unit **35b** held by the second support stand **175b** from the second driving position Pib in the apparatus casing **4** to the second maintenance position Pob outside the apparatus casing **4**, and push them back again to the second driving position Pib.

[0054] Further, in the second slider **41b**, the second detection unit **45b** including, for example, a photo-microsensor is fixed. Moreover, in the second support stand **175b**, a sensor dog (not shown) for position detection is fixed. The second detection unit **45b** outputs a detection signal each time the second detection unit **45b** detects the sensor dog that passes in the vicinity as the second support stand **175b** moves along the X direction. Hereinafter, the detection signal output from the second detection unit **45b** is referred to as a “second detection signal”. That is, the second detection unit **45b** detects movement of the second head unit **35b** being supported by the second support stand **175b** between the second driving position Pib and the second maintenance position Pob, and outputs the second detection signal. Thus, the control unit **9** described later can grasp a position of the second head unit **35b** by referring to the second detection signal output from the second detection unit **45b**.

[0055] Likewise, in the third slider **41c**, the third detection unit **45c** is fixed. The third detection unit **45c** detects movement of the third head unit **35c** being supported by the support stand **175** between the third driving position Pic and the third maintenance position Poc, and outputs a third detection signal. Likewise, in the fourth slider **41d**, the fourth detection unit **45d** is fixed. The fourth detection unit **45d** detects movement of the fourth head unit **35d** being supported by the support stand **175** between the fourth driving position Pid and the fourth maintenance position Pod, and outputs a fourth detection signal.

[0056] Next, a configuration of the ink supply unit **36** is described in detail. The ink supply unit **36** is a device configured to supply ink to the head unit **35** while circulating a part of ink. As shown in FIG. **3**, the ink supply unit **36** includes a supply tank **51**, a collecting tank **52**, a supply-side manifold **61**, and a plurality of supply-side narrow pipes **62**, a plurality of collecting-side narrow pipes **63**, a collecting-side manifold **64**, a feedback pipe **65**, a circulation pump **71**, a plurality of supply-side on-off valves **73**, a plurality of head outlet-side on-off valves **74**, a heater **76**, a filter **87**, and a deaeration unit **88**. In the present preferred embodiment, the ink supply unit **36** includes five supply-side narrow pipes **62**, five collecting-side narrow pipes **63**, five supply-side on-off valves **73**, and five head outlet-side on-off valves **74**.

[0057] The supply tank **51** is a container for temporally storing ink to be supplied to the discharge head **811**. In the supply tank **51**, an internal chamber **510** in which ink can be temporally stored is provided.

[0058] The supply-side manifold **61** and the five supply-side narrow pipes **62** are pipes connecting the supply tank **51** and the five discharge heads **811** included in one head unit **35**. The supply-side manifold **61** is a wide pipe having an upstream end that is connected so as to communicate with the internal chamber **510** of the supply tank **51**. Each of the five supply-side narrow pipes **62** is a narrow pipe branching from the supply-side manifold **61**. Each of the five supply-side narrow pipes **62** has an upstream end communicating with an internal passage of the supply-side manifold **61**, and has a downstream end that is connected so as to communicate with the internal tank **102** of one discharge head **811**.

[0059] Further, in the present preferred embodiment, the supply-side on-off valve **73** is interposed in each of the supply-side narrow pipes **62**. For the supply-side on-off valve **73**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used.

Alternatively, for the supply-side on-off valve **73**, an on-off valve that is manually opened and closed may be used. While the supply-side on-off valve **73** is closed, communication of an internal passage of the supply-side narrow pipe **62** is blocked. That is, while the supply-side on-off valve **73** is closed, flow of ink from the supply tank **51** to the discharge head **811** is interrupted. Meanwhile, while the supply-side on-off valve **73** is opened, communication of the internal passage of the supply-side narrow pipe **62** is allowed. Note that the supply-side on-off valve **73** is not necessarily required to be provided.

[0060] The five collecting-side narrow pipes **63** and the collecting-side manifold **64** are pipes connecting the five discharge heads **811** included in the head unit **35** and the collecting tank **52**. Each of the five collecting-side narrow pipes **63** is a narrow pipe branching from the collecting-side manifold **64**. Each of the five collecting-side narrow pipes **63** has an upstream end that is connected so as to communicate with the internal tank **102** of one discharge head **811**, and has a downstream end that is connected so as to communicate with an internal passage of the collecting-side manifold **64**. The collecting-side manifold **64** is a wide pipe having a downstream end that is connected so as to communicate with an internal chamber **520** of the collecting tank **52** described later.

[0061] Further, in the present preferred embodiment, the head outlet-side on-off valve **74** is interposed in each of the collecting-side narrow pipes **63**. For the head outlet-side on-off valve **74**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used. Alternatively, for the head outlet-side on-off valve **74**, an on-off valve that is manually opened and closed may be used. While the head outlet-side on-off valve **74** is closed, communication of an internal passage of the collecting-side narrow pipe **63** is blocked. That is, while the head outlet-side on-off valve **74** is closed, flow of ink from the discharge head **811** to the collecting tank **52** is interrupted. Meanwhile, while the head outlet-side on-off valve **74** is opened, communication of the internal passage of the collecting-side narrow pipe **63** is allowed. Note that the head outlet-side on-off valve **74** is not necessarily required to be provided.

[0062] The collecting tank **52** is a container for temporally storing ink collected from the discharge heads **811**. In the collecting tank **52**, the internal chamber **520** in which ink can be temporally

stored is provided.

[0063] Further, as shown in FIG. 3, the supply tank **51** is connected to a pressurization mechanism **515**. The pressurization mechanism **515** pressurizes the inside of the supply tank **51**, to regulate a pressure of the internal chamber **510** of the supply tank **51** to a positive pressure. Specifically, the pressurization mechanism **515** regulates a pressure of the internal chamber **510** of the supply tank **51** to a pressure higher than the atmospheric pressure. The pressurization mechanism **515** includes, for example, a compressor, a pressurization buffer tank, a pressure regulation mechanism (regulator), and the like.

[0064] Meanwhile, the collecting tank **52** is connected to a decompression mechanism **524**. The decompression mechanism **524** decompresses the inside of the collecting tank **52**, to regulate a pressure of the internal chamber **520** of the collecting tank **52** to a negative pressure. Specifically, the decompression mechanism **524** regulates a pressure of the internal chamber **520** of the collecting tank **52** to a pressure lower than the atmospheric pressure. The decompression mechanism **524** includes, for example, a vacuum pump, a decompression buffer tank, a pressure regulation mechanism (regulator), and the like.

[0065] Note that the pressurization mechanism **515** and the decompression mechanism **524** are configured such that the operations thereof can be controlled by the control unit **9**. When the pressurization mechanism **515** and the decompression mechanism **524** are driven, there is generated a pressure difference between the internal chamber **510** of the supply tank **51** and the internal chamber **520** of the collecting tank **52**. Thus, ink stored in the supply tank **51** can be supplied to each discharge head **811**, and further, ink remaining in each discharge head **811** can be collected into the collecting tank **52**. Note that the ink remaining in the discharge head **811** is ink being left undischarged in each discharge head **811**.

[0066] The feedback pipe **65** is a pipe connecting the internal chamber **520** of the collecting tank **52** and the internal chamber **510** of the supply tank **51** such that the internal chambers can communicate with each other. In other words, the feedback pipe **65** connects the collecting tank **52** and the supply tank **51**. As shown in FIG. 3, an internal passage of the feedback pipe **65** has an upstream end that is connected so as to communicate with the internal chamber **520** of the collecting tank **52**. Further, the internal passage of the feedback pipe **65** has a downstream end that is connected so as to communicate with the internal chamber **510** of the supply tank **51**.

[0067] With the above-described configuration, there is formed an ink circulation path **55** that extends from the supply tank **51**, passes through the supply-side manifold **61**, the supply-side narrow pipes **62**, the internal tanks **102** of the discharge heads **811**, the collecting-side narrow pipes **63**, the collecting-side manifold **64**, the collecting tank **52**, and the feedback pipe **65**, and returns back to the supply tank **51**. In other words, the circulation path **55** includes the supply tank **51**, the discharge heads **811**, the collecting tank **52**, and the feedback pipe **65**. Further, in the feedback pipe **65**, the circulation pump **71**, the heater **76**, the filter **87**, and the deaeration unit **88** are interposed.

[0068] The circulation pump **71** is a device configured to perform a liquid-delivery operation of delivering ink from the collecting tank **52** to the supply tank **51** via the feedback pipe **65**. The circulation pump **71** is electrically connected to the control unit **9**. The circulation pump **71** generates flow of ink from the collecting tank **52** to the supply tank **51** in the internal passage of the feedback pipe **65** in response to an operation signal from the control unit **9**.

[0069] The heater **76** is a device configured to heat ink delivered through the internal passage of the feedback pipe **65**. The heater **76** heats ink flowing from the collecting tank **52** to the supply tank **51**. The heater **76** is positioned between the circulation pump **71** and the supply tank **51** in the feedback pipe **65**. The heater **76** is electrically connected to the control unit **9**.

[0070] The filter **87** is interposed on the downstream side of the heater **76** with respect to an ink delivery direction and on the upstream side of the supply tank **51** with respect to the ink delivery direction, in the feedback pipe **65**. The filter **87** filters ink delivered through the internal passage of the feedback pipe **65**, to remove foreign matters included in the ink.

[0071] The deaeration unit **88** is interposed on the downstream side of the filter **87** with respect to the ink delivery direction and on the upstream side of the supply tank **51** with respect to the ink delivery direction, in the feedback pipe **65**. The deaeration unit **88** of the present preferred embodiment is a so-called hollow-fiber membrane deaeration module. The deaeration unit **88** removes bubbles in ink delivered through the internal passage of the feedback pipe **65**.

[0072] Next, the control unit **9** is described. The control unit **9** is an information processing device configured to control each component of the inkjet printing apparatus **1**. FIG. **8** is a block diagram showing connection between the control unit **9** and each component of the inkjet printing apparatus **1**. As conceptually shown in FIG. **8**, the control unit **9** includes a processor **91** such as a CPU, a memory **92** such as a RAM, and a storage unit **93** such as a hard disk drive. In the storage unit **93**, a computer program **9P** for conveying the continuous paper, **10**, performing printing, circulating ink, performing maintenance of the first head unit **35a** to the fourth head unit **35d**, is stored.

[0073] Further, as shown in FIG. **8**, the control unit **9** is electrically connected to the conveyor unit **2**, to the discharge heads **811**, the main movement mechanism **813**, and the subsidiary movement mechanism **815** of each of the four head units **35a** to **35d** of the printing unit **3**, to the circulation pump **71**, the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, the heater **76**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **36a** to **36d** of the printing unit **3**, to the four sliders **41a** to **41d**, and to the four detection units **45a** to **45d** such that the control unit **9** can conduct communication to/from each component. The control unit **9** controls operations of those components in accordance with the computer program **9P**.

[0074] Further, as shown in FIG. **8**, the control unit **9** is electrically connected to the input unit **11** such that the control unit **9** can conduct communication to/from the input unit **11**. The input unit **11** is a device that can receive a command for making a transition of the plurality of head units **35** to a maintenance mode. More specifically, the input unit **11** is a device that can receive a first command for moving the cap **812** of the first head unit **35a** from the facing position **P1** to the receding position **P2**, and/or receive a second command for moving the cap **812** of the second head unit **35b** from the facing position **P1** to the receding position **P2**, and/or receive a third command for moving the cap **812** of the third head unit **35c** from the facing position **P1** to the receding position **P2**, and/or receive a fourth command for moving the cap **812** of the fourth head unit **35d** from the facing position **P1** to the receding position **P2**.

[0075] Note that the first command includes a command for driving the subsidiary movement mechanism **815** of the first head unit **35a** and switching the ink discharge surface **161** of each of the five discharge heads **811** from a “capped state” to a “decapped state”. The second command includes a command for driving the subsidiary movement mechanism **815** of the second head unit **35b** and switching the ink discharge surface **161** of each of the five discharge heads **811** from a “capped state” to a “decapped state”. The third command includes a command for driving the subsidiary movement mechanism **815** of the third head unit **35c** and switching the ink discharge surface **161** of each of the five discharge heads **811** from a “capped state” to a “decapped state”. The fourth command includes a command for driving the subsidiary movement mechanism **815** of the fourth head unit **35d** and switching the ink discharge surface **161** of each of the five discharge heads **811** from a “capped state” to a “decapped state”.

[0076] The input unit **11** includes, for example, an input interface such as a touch panel. The operator individually inputs, to the input unit **11**, the above-described first to fourth commands for making a transition of the first head unit **35a** to fourth head unit **35d** to a maintenance mode via the input interface. Then, upon receipt of the first command, the input unit **11** outputs a first input signal which is a signal related to the first command, to the control unit **9**. Upon receipt of the second command, the input unit **11** outputs a second input signal which is a signal related to the second command, to the control unit **9**. Upon receipt of the third command, the input unit **11** outputs a third input signal which is a signal related to the third command, to the control unit **9**.

Upon receipt of the fourth command, the input unit **11** outputs a fourth input signal which is a signal related to the fourth command, to the control unit **9**. Meanwhile, the input unit **11** may be formed integrally with the control unit **9** or another unit.

2. Procedure for Conveying Continuous Paper, Printing, Circulating Ink, and Performing Maintenance of Head Unit

[0077] Next, a procedure for conveying the continuous paper **10**, performing printing on the continuous paper **10**, circulating ink, and performing maintenance of the first head unit **35a** to the fourth head unit **35d** performed in the inkjet printing apparatus **1** is described.

[0078] In conveying the continuous paper **10**, performing printing on the continuous paper **10**, and circulating ink, first, the control unit **9** causes the conveyor unit **2** to operate, to thereby convey the continuous paper **10** along a predetermined conveying path in the length direction. Then, the control unit **9** controls the plurality of nozzles **103** of each of the four head units **35** (the first head unit **35a** to the fourth head unit **35d**) while conveying the continuous paper **10**, to discharge droplets of the first ink to the fourth ink onto the surface of the continuous paper **10**. As a result of this, a multicolor image is recorded on the surface of the continuous paper **10**.

[0079] Meanwhile, as advance preparation for conveying the continuous paper **10**, performing printing on the continuous paper **10**, and circulating ink, a sufficient amount of ink is stored in the internal chamber **510** of the supply tank **51** of each of the four ink supply units **36** (the first ink supply unit **36a** to the fourth ink supply unit **36d**). Further, the control unit **9** opens the five supply-side on-off valves **73** and the five head outlet-side on-off valves **74** of each ink supply unit **36**. Then, the control unit **9** drives the circulation pump **71**, the heater **76**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **36** (the first ink supply unit **36a** to the fourth ink supply unit **36d**). Specifically, the control unit **9** drives the circulation pump **71** to circulate ink through the ink circulation path **55** while driving the pressurization mechanism **515** and the decompression mechanism **524**, to thereby supply the first ink to the fourth ink to the internal tanks **102** of the corresponding discharge heads **811**, respectively.

[0080] Next, a procedure for performing maintenance of the first head unit **35a** to the fourth head unit **35d** is described. FIG. **9** is a flowchart showing the procedure for performing maintenance of the first head unit **35a** to the fourth head unit **35d**. It is supposed that the inkjet printing apparatus **1** is in a standby state at a time of starting maintenance shown in FIG. **9**. Specifically, conveyance of the continuous paper **10** is stopped, and the first head unit **35a** to the fourth head unit **35d** stop discharging ink droplets onto the continuous paper **10**. Further, the cap **812** of each of the first head unit **35a** to the fourth head unit **35d** is positioned at the facing position **P1**. Moreover, the five discharge heads **811** of each of the first head unit **35a** to the fourth head unit **35d** are moved down by driving of the subsidiary movement mechanism **815**. As a result of this, the ink discharge surface **161** of each of the discharge heads **811** is isolated and enclosed from the outside atmosphere with the cap **812**, and thus is in a “capped state”

[0081] While the inkjet printing apparatus is in the standby state described above, maintenance of the first head unit **35a** to the fourth head unit **35d** is started. The operator inputs a command for making a transition of any or all of the first head unit **35a** to the fourth head unit **35d** to a maintenance mode via the input interface of the input unit **11** (step **S1**). In other words, the operator inputs the first command, the second command, the third command, and/or the fourth command described above via the input interface of the input unit **11**. In the following description, there is supposed a case in which the operator inputs a command (the first command and the second command) for performing maintenance of the first head unit **35a** and the second head unit **35b**, as an example.

[0082] In this case, the input unit **11** outputs the first input signal to the control unit **9** upon receipt of the first command. Further, the input unit **11** outputs the second input signal to the control unit **9** upon receipt of the second command. Thus, the control unit **9** receives the first input signal and the

second input signal from the input unit **11**. However, at this stage, the control unit **9** does not perform control for driving the main movement mechanism **813** and the subsidiary movement mechanism **815** of each of the first head unit **35a** and the second head unit **35b**.

[0083] Subsequently, the operator pulls out any of the first head unit **35a** to the fourth head unit **35d** to the outside space of the apparatus casing **4** to perform maintenance (step S2). For example, the operator first pulls out the first support stand **175a** and the first head unit **35a** placed on the first slider **41a** from the first driving position Pia in the apparatus casing **4** to the first maintenance position Poa outside the apparatus casing **4**. Then, the first detection unit **45a** detects that the first head unit **35a** has moved between the first driving position Pia and the first maintenance position Poa, and outputs the first detection signal. Thus, the control unit **9** can grasp that the first head unit **35a** has moved from the first driving position Pia to the first maintenance position Poa by referring to the first detection signal output from the first detection unit **45a**.

[0084] Meanwhile, the control unit **9** stops driving the circulation pump **71** of the first ink supply unit **36a** when it is detected that the first head unit **35a** has moved from the first driving position Pia to the first maintenance position Poa. By stopping circulation of the first ink, it is possible to prevent ink leakage from the discharge ports **151** of the discharge heads **811** of the first head unit **35a**, intake of air from the outside atmosphere through the discharge ports **151**, and the like even though the first head unit **35a** has been pulled out to the outside of the apparatus casing **4**. This can suppress a change in a pressure applied to the discharge heads **811**, to thereby prevent damage or the like to the discharge heads **811**. Further, the control unit **9** closes the supply-side on-off valves **73** and the head outlet-side on-off valves **74** of the first ink supply unit **36a**. This can further prevent ink leakage from the discharge ports **151** of the discharge heads **811** of the first head unit **35a**, intake of air from the outside atmosphere through the discharge ports **151**, and the like.

[0085] Subsequently, the control unit **9** checks whether the head unit **35** having been pulled out to the outside space of the apparatus casing **4** has been input to the input unit **11**, as an object of maintenance in the step S1, in the first place (step S3). In the present preferred embodiment, it is checked whether the first head unit **35a** having moved to the first maintenance position Poa outside the apparatus casing **4** has been input to the input unit **11**, as an object of maintenance. The first head unit **35a** has been input to the input unit **11**, as an object of maintenance in the step S1 as described above (step S3: YES), and thus the control unit **9** drives the main movement mechanism **813** and the subsidiary movement mechanism **815** of the first head unit **35a** after a certain time lag. More specifically, the control unit **9** drives the subsidiary movement mechanism **815** of the first head unit **35a**, and switches the ink discharge surface **161** of each discharge head **811** of the first head unit **35a** to a “decapped state” (step S4). This allows the operator to check the condition of the ink discharge surface **161** of each discharge head **811** and perform a part of maintenance. After that, the control unit **9** drives the main movement mechanism **813**, to move the cap **812** of the first head unit **35a** from the facing position P1 to the receding position P2, and entirely exposes the ink discharge surface **161** of each discharge head **811** (step S5). This allows the operator to make full access to the ink discharge surface **161** and thus easily perform cleaning of portions near the nozzles **103**, part replacement, and the like.

[0086] Then, when finishing maintenance including cleaning of each discharge head **811**, part replacement, and the like, the operator pushes back the first support stand **175a** and the first head unit **35a** placed on the first slider **41a** from the first maintenance position Poa outside the apparatus casing **4** to the first driving position Pia in the apparatus casing **4**. As a result, the first detection unit **45a** detects that the first head unit **35a** has moved between the first maintenance position Poa and the first driving position Pia, and outputs the first detection signal. Thus, the control unit **9** can grasp that the first head unit **35a** has moved from the first maintenance position Poa to the first driving position Pia by referring to the first detection signal output from the first detection unit **45a**. Meanwhile, when it is detected that the first head unit **35a** has moved from the first maintenance position Poa to the first driving position Pia (step S6: YES), the control unit **9** drives the main

movement mechanism **813** of the first head unit **35a** again. Thus, the control unit **9** moves the cap **812** from the receding position **P2** to the facing position **P1**, so that the cap **812** faces each discharge head **811** (step **S7**). Further, the control unit **9** drives the subsidiary movement mechanism **815** of the first head unit **35a** again. As a result, the head holding plate **814** and the five discharge heads **811** of the first head unit **35a** are moved down. Consequently, the ink discharge surface **161** of each discharge head **811** can be switched again to a “capped state” in which the ink discharge surface **161** is isolated and enclosed from the outside atmosphere with the cap **812** (step **S8**).

[0087] As described above, the control unit **9** can perform operations of pulling out the apparatus casing **4** to the outside space, performing maintenance including cleaning, part replacement, and the like, and pushing back the apparatus casing **4** to the inside space, on each of the first head unit **35a** to the fourth head unit **35d**. Then, the cap **812** in each of the head units **35a** to **35d** can be moved to the receding position **P2**, and respective commands for switching the head units **35a** to **35d** to a “decapped state” can be input at once in the input unit **11**. This can reduce a burden on the operator. Further, the ink discharge surfaces **161** of the five discharge heads **811** of the head units **35a** to **35d** to be subjected to maintenance are switched to a “decapped state” immediately before maintenance of the head units **35a** to **35d** is actually performed. Thus, a time period for which the ink discharge surfaces **161** of the head units **35a** to **35d** to be subjected to maintenance are exposed to the outside atmosphere can be further shortened. Consequently, the nozzles **103** formed in the ink discharge surfaces **161** can be prevented from being dried. Meanwhile, the ink discharge surface **161** of the head unit **35** not to be subjected to maintenance is kept in a “capped state” also during maintenance of another head unit **35**. In the case described in the present preferred embodiment, the ink discharge surfaces **161** of the third head unit **35c** and the fourth head unit **35d** are kept in a “capped state” also during maintenance of the first head unit **35a** and the second head unit **35b**. Therefore, the nozzles **103** formed in the ink discharge surfaces **161** of the head units **35** not to be subjected to maintenance can be prevented from being dried.

[0088] In other words, the control unit **9** of the present preferred embodiment can perform each of a step a) of controlling the subsidiary movement mechanism **815** (first movement mechanism) of the first head unit **35a**, to switch the ink discharge surfaces **161** of the first head unit **35a** from a “capped state” to a “decapped state” after elapse of a first predetermined time period when it is detected on the basis of the first detection signal received from the first detection unit **45a**, that the first head unit **35a** has moved from the first driving position **Pia** to the first maintenance position **Poa** while the first input signal has been received from the input unit **11**; and a step b) of controlling the subsidiary movement mechanism **815** (second movement mechanism) of the second head unit **35b**, to switch the ink discharge surfaces **161** of the second head unit **35b** from a “capped state” to a “decapped state” after elapse of a second predetermined time period when it is detected on the basis of the second detection signal received from the second detection unit **45b**, that the second head unit **35b** has moved from the second driving position **Pib** to the second maintenance position **Pob** while the second input signal has been received from the input unit **11**.

[0089] Further, as described above, when it is detected that any of the head units **35** has moved from the driving position in the apparatus casing **4** to the maintenance position outside the apparatus casing **4**, the control unit **9** stops driving the circulation pump **71** of the ink supply unit **36** corresponding to the head unit **35** having moved. This results in stopping circulation of ink in the ink supply unit **36** corresponding to the head unit **35** having moved to the maintenance position. Consequently, even though the head unit **35** has been pulled out to the outside of the apparatus casing **4**, ink leakage from the discharge ports **151** of the discharge heads **811** of the pulled-out head unit **35**, intake of air from the outside atmosphere through the discharge ports **151**, and the like can be prevented. This can suppress a change in a pressure applied to the discharge heads **811**, to thereby prevent damage or the like to the discharge heads **811**. Further, in the ink supply unit **36** corresponding to the head unit **35** not to be subjected to maintenance, ink circulation is continued

also during maintenance of another head unit **35**. In the case described in the present preferred embodiment, in the ink supply units **36** corresponding to the third head unit **35c** and the fourth head unit **35d**, ink circulation is continued also during maintenance of the first head unit **35a** and the second head unit **35b**. Therefore, the discharge ports **151** of the discharge heads **811** of the head units **35** not to be subjected to maintenance can be further prevented from being dried.

[0090] Specifically, when it is detected that the first head unit **35a** has moved from the first driving position *Pia* to the first maintenance position *Poa*, the control unit **9** of the present preferred embodiment stops driving the circulation pump **71** (first circulation pump) of the first ink supply unit **36a**. Meanwhile, when it is detected that the second head unit **35b** has moved from the second driving position *Pib* to the second maintenance position *Pob*, the control unit **9** of the present preferred embodiment stops driving the circulation pump **71** (second circulation pump) of the second ink supply unit **36b**. In this manner, in the present preferred embodiment, ink circulation is stopped immediately before maintenance of the head units **35a** and **35b**, and thus the discharge ports **151** can be further prevented from being dried.

[0091] Further, as described above, when finishing maintenance including cleaning, part replacement, and the like of any of the head units **35a** to **35d**, the operator pushes back the head unit **35a**, **35b**, **35c**, or **35d** from the maintenance position *Poa*, *Pob*, *Poc*, or *Pod* outside the apparatus casing **4** to the driving position *Pia*, *Pib*, *Pic*, or *Pid* in the apparatus casing **4**. Then, when it is detected on the basis of the first, second, third, or fourth detection signal, that any of the head units **35a** to **35d** has moved from the maintenance position *Poa*, *Pob*, *Poc*, or *Pod* to the driving position *Pia*, *Pib*, *Pic*, or *Pid*, the control unit **9** drives again the main movement mechanism **813** and the subsidiary movement mechanism **815** of the corresponding head unit **35a**, **35b**, **35c**, or **35d**. Thus, the control unit **9** moves the cap **812** of the corresponding head unit **35a**, **35b**, **35c**, or **35d** from the receding position *P2* to the facing position *P1*. Further, the control unit **9** brings the ink discharge surfaces **161** of the discharge heads **811** of the corresponding head unit **35a**, **35b**, **35c**, or **35d** into contact with the corresponding cap **812** again. As a result, the ink discharge surfaces **161** are switched again to a “capped state” by the cap **812** immediately after an end of maintenance of the corresponding head unit **35a**, **35b**, **35c**, or **35d**. Hence, a time period for which the ink discharge surfaces **161** of the corresponding head unit **35a**, **35b**, **35c**, or **35d** having been subjected to maintenance are exposed to the outside atmosphere can be further shortened. Consequently, the ink discharge surfaces **161**, together with the nozzles **103** formed in the ink discharge surfaces **161**, can be further prevented from being dried.

[0092] In other words, after the above-described step a), when it is detected on the basis of the first detection signal further received from the first detection unit **45a**, that the first head unit **35a** has moved from the first maintenance position *Poa* to the first driving position *Pia*, the control unit **9** controls the subsidiary movement mechanism **815** (first movement mechanism) of the first head unit **35a**, to cap the ink discharge surface **161** (first ink discharge surface) of each discharge head **811** of the first head unit **35a** with the corresponding cap **812** (first cap). Meanwhile, after the above-described step b), when it is detected on the basis of the second detection signal further received from the second detection unit **45b**, that the second head unit **35b** has moved from the second maintenance position *Pob* to the second driving position *Pib*, the control unit **9** controls the subsidiary movement mechanism **815** (second movement mechanism) of the second head unit **35b**, to cap the ink discharge surface **161** (second ink discharge surface) of each discharge head **811** of the second head unit **35b** with the corresponding cap **812** (second cap).

[0093] Moreover, as described above, in the step **S3**, the control unit **9** checks whether the head unit **35** having been pulled out to the outside space of the apparatus casing **4** has been input to the input unit **11**, as an object of maintenance in the step **S1**, in the first place. Then, the control unit **9** opens or closes the cap **812** only when an answer to the step **S3** is YES. With this configuration, the operator can proceed with maintenance work while checking whether the maintenance work are performed as originally planned. This results in performing the work more correctly.

[0094] Here, suppose a case in which maintenance of the first head unit **35a** is not scheduled and the operator does not input the first command to the input unit **11** in the step **S1**. Then, when it is checked that the first head unit **35a** having been pulled out to the outside of the apparatus casing **4**, has not been input to the input unit **11** as an object of maintenance in the step **S3** (step **S3**: NO), the control unit **9** recognizes the presence of an error. In this case, the control unit **9** waits without driving the main movement mechanism **813** and the subsidiary movement mechanism **815** of the first head unit **35a**. Further, the control unit **9** displays an error message on a display (not shown) or the like provided in the input unit **11**, or generates an alarm or the like, to perform error notification (step **S9**).

[0095] Specifically, when it is detected on the basis of the first detection signal received from the first detection unit **45a**, that the first head unit **35a** has moved from the first driving position **Pia** to the first maintenance position **Poa** while the first input signal is not received from the input unit **11**, the control unit **9** of the present preferred embodiment recognizes the presence of an error. Then, the control unit **9** waits while keeping the ink discharge surface **161** (first ink discharge surface) of each discharge head **811** of the first head unit **35a** in a “capped state” without performing control of the main movement mechanism **813** and the subsidiary movement mechanism **815** of the first head unit **35a**. Meanwhile, when it is detected on the basis of the second detection signal received from the second detection unit **45b**, that the second head unit **35b** has moved from the second driving position **Pib** to the second maintenance position **Pob** while the second input signal is not received from the input unit **11**, the control unit **9** of the present preferred embodiment recognizes the presence of an error. Then, the control unit **9** waits while keeping the ink discharge surface **161** (second ink discharge surface) of each discharge head **811** of the second head unit **35b** in a “capped state” without performing control of the main movement mechanism **813** and the subsidiary movement mechanism **815** of the second head unit **35b**. With this configuration, in a case in which the head unit **35** that is not scheduled to be pulled out to the outside space is pulled out, the operator can recognize the presence of an error and thus can discontinue the rest of the work.

3. Modifications

[0096] Hereinabove, the present preferred embodiment of the present invention has been described, but the present invention is not limited to the above-described preferred embodiment.

[0097] In the above-described preferred embodiment, the subsidiary movement mechanism **815** of the head unit **35** is driven, and the head holding plate **814** and each discharge head **811** are moved up/down. Thus, switching of the ink discharge surface **161** of each discharge head **811** between a “capped state” and a “decapped state” is achieved. As an alternative to upward/downward movement of each discharge head **811** of the head unit **35**, the cap **812** may be moved up/down in order to achieve switching of the ink discharge surface **161** of each discharge head **811** between a “capped state” and a “decapped state”

[0098] In the above-described preferred embodiment, by driving of the subsidiary movement mechanism **815** of the head unit **35**, switching of the ink discharge surface **161** of each discharge head **811** between a “capped state” and a “decapped state” is achieved. Further, by driving of the main movement mechanism **813**, the cap **812** is moved along the X direction. As a result of this, the ink discharge surface **161** of each discharge head **811** is entirely exposed from the cap **812**. However, the main movement mechanism **813** may be omitted on condition that the operator can make full access to the ink discharge surface **161** and perform various kinds of maintenance work only by changing a distance between the ink discharge surface **161** and the cap **812** along the Z direction.

[0099] In the above-described preferred embodiment, the operator gives a command via the input unit **11** to the effect that switching of the ink discharge surface **161** of each discharge head **811** between a “capped state” and a “decapped state” by driving of the subsidiary movement mechanism **815** of the head unit **35**, and movement of the cap **812** in a “decapped state” between the facing position **P1** and the receding position **P2** are successively performed. Alternatively,

switching of the ink discharge surface **161** between a “capped state” and a “decapped state” and movement of the cap **812** between the facing position **P1** and the receding position **P2** may be individually requested. That is, the operator may be allowed to give a command for switching the ink discharge surface **161** from a “capped state” to a “decapped state” and a command for moving the cap **812** to the receding position **P2** after decapping, individually, via the input unit **11**.

[0100] Likewise, the operator may be allowed to give a command for moving the cap **812** of the head unit **35** from the receding position **P2** to the facing position **P1** and a command to the cap **812** at the facing position **P1** for placing the ink discharge surface **161** of each discharge head **811** into a “capped state”, individually, via the input unit **11**.

[0101] Further, the respective elements described in the above-described embodiment may be appropriately combined unless contradiction occurs.

[0102] While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

Claims

1. An inkjet printing apparatus that performs printing on a printing medium by an inkjet method, comprising: an apparatus casing; a first head unit including a first discharge head having a first ink discharge surface in which a first discharge port facing down, through which first ink is discharged, is formed, a first cap configured to cap the first ink discharge surface, and a first movement mechanism configured to switch the first ink discharge surface between a capped state in which the first ink discharge surface is isolated from an outside atmosphere with the first cap and a decapped state in which the capped state is canceled; a second head unit including a second discharge head having a second discharge surface in which a second discharge port facing down, through which second ink is discharged, is formed, a second cap configured to cap the second ink discharge surface, and a second movement mechanism configured to switch the second ink discharge surface between a capped state in which the second ink discharge surface is isolated from the outside atmosphere with the second cap and a decapped state in which the capped state is canceled; a first slider supporting the first head unit such that the first head unit is slidable between a first driving position in an inside space of the apparatus casing and a first maintenance position in an outside space of the apparatus casing; a second slider that is provided separately from the first slider and supports the second head unit such that the second head unit is slidable between a second driving position in the inside space of the apparatus casing and a second maintenance position in the outside space of the apparatus casing; a first detection unit configured to detect movement of the first head unit between the first driving position and the first maintenance position and output a first detection signal; a second detection unit configured to detect movement of the second head unit between the second driving position and the second maintenance position and output a second detection signal; an input unit configured to receive a first command including a command for switching the first ink discharge surface from the capped state to the decapped state and output a first input signal upon receipt of the first command, the input unit being configured to receive a second command including a command for switching the second ink discharge surface from the capped state to the decapped state and output a second input signal upon receipt of the second command; and a control unit electrically connected to the first discharge head, the first movement mechanism, the first detection unit, the second discharge head, the second movement mechanism, the second detection unit, and the input unit, wherein the control unit is capable of individually performing a step a) of controlling the first movement mechanism so as to switch the first ink discharge surface from the capped state to the decapped state after elapse of a first predetermined time period when it is detected on the basis of the first detection signal received from the first detection unit, that the first head unit has moved from the first driving position to the first

maintenance position while the first input signal has been received from the input unit, and a step b) of controlling the second movement mechanism so as to switch the second ink discharge surface from the capped state to the decapped state after elapse of a second predetermined time period when it is detected on the basis of the second detection signal received from the second detection unit, that the second head unit has moved from the second driving position to the second maintenance position while the second input signal has been received from the input unit.

2. The inkjet printing apparatus according to claim 1, wherein after the step a), the control unit controls the first movement mechanism so as to switch the first ink discharge surface from the decapped state to the capped state when it is detected on the basis of the first detection signal further received from the first detection unit, that the first head unit has moved from the first maintenance position to the first driving position, and after the step b), the control unit controls the second movement mechanism so as to switch the second ink discharge surface from the decapped state to the capped state when it is detected on the basis of the second detection signal further received from the second detection unit, that the second head unit has moved from the second maintenance position to the second driving position.

3. The inkjet printing apparatus according to claim 1, wherein the control unit recognizes presence of an error and waits while keeping the first ink discharge surface in the capped state when it is detected on the basis of the first detection signal received from the first detection unit, that the first head unit has moved from the first driving position to the first maintenance position while the first input signal is not received from the input unit, and the control unit recognizes presence of an error and waits while keeping the second ink discharge surface in the capped state when it is detected on the basis of the second detection signal received from the second detection unit, that the second head unit has moved from the second driving position to the second maintenance position while the second input signal is not received from the input unit.

4. The inkjet printing apparatus according to claim 1, further comprising: a first circulation path including a first supply tank in which the first ink supplied to the first discharge head is stored, a first collecting tank in which the first ink collected from the first discharge head is stored, and a first feedback pipe connecting the first collecting tank and the first supply tank; a first circulation pump interposed in the first feedback pipe, the first circulation pump being configured to deliver the first ink from the first collecting tank to the first supply tank via the first feedback pipe; a second circulation path including a second supply tank in which the second ink supplied to the second discharge head is stored, a second collecting tank in which the second ink collected from the second discharge head is stored, and a second feedback pipe connecting the second collecting tank and the second supply tank; and a second circulation pump interposed in the second feedback pipe, the second circulation pump being configured to deliver the second ink from the second collecting tank to the second supply tank via the second feedback pipe, wherein the control unit is further electrically connected to the first circulation pump and the second circulation pump, the control unit stops driving the first circulation pump when it is detected that the first head unit has moved from the first driving position to the first maintenance position, and the control unit stops driving the second circulation pump when it is detected that the second head unit has moved from the second driving position to the second maintenance position.
